

AN AUTOMATED STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE WITH OPTIONAL JUMPS

RADU BRICIU

ABSTRACT. We specify a routine that calibrates the five canonical Heston [1993] parameters $\Phi = (\theta, \kappa, \eta, \rho, v_0)$ to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot S_{ref} , the moneyness-normalised and grid-snapped strike K^* , the surface-coverage requirements, the configurable in-engine objective (defaulting to implied-volatility “IV-space” residuals), the IV-space restart selection and acceptance gate, and the boundary-pegging rejection rule. The same routine optionally fits the Bates [1996] extension, which adds Merton lognormal jumps to the Heston dynamics, selected by a run-time flag and calibrated over the identical surface and gate. We keep Heston as the default throughout and present Bates as the eight-parameter generalisation it specialises from.

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1. The model

1.1. Basis. Under the risk-neutral measure, with constant risk-free rate r and continuous dividend yield g , the log-price $X_t = \ln S_t$ and instantaneous variance v_t of the SPX follow the Heston [1993] dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where W, B are independent Brownian motions, so $\text{corr}(dX_t, dv_t) = \rho dt$. The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with $\theta > 0$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\eta > 0$ the volatility of variance (QuantLib's `sigma`), $\rho \in [-1, 1]$ the leverage correlation, and $v_0 > 0$ the initial variance. The Feller quantity [Feller, 1951]

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit; $\mathcal{F}(\Phi) \geq 0$ guarantees $v_t > 0$ almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

1.2. Jumps. The Bates [1996] model extends Heston with Merton [1976] log-normal jumps in the log-price. The variance dynamics (2) are unchanged. The level equation (1) gains a compound-Poisson jump term:

$$dX_t = (r - g - \frac{1}{2}v_t - \lambda\xi) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t) + dJ_t. \quad (5)$$

Here $J_t = \sum_{n=1}^{N_t} Y_n$ is a compound-Poisson process: N_t is a Poisson process of intensity $\lambda \geq 0$ (expected jumps per year), independent of W and B , and the log-jump sizes $Y_n \sim \mathcal{N}(\nu, \delta^2)$ are i.i.d. The compensator

$$\xi = \mathbb{E}[e^Y - 1] = e^{\nu + \delta^2/2} - 1 \quad (6)$$

offsets the jump drift so the discounted price stays a martingale, leaving the level drift $r - g$ in expectation. The augmented parameter vector is

$$\Theta = (\theta, \kappa, \eta, \rho, v_0, \lambda, \nu, \delta) \in \mathbb{R}^8, \quad (7)$$

the five Heston parameters of (1)–(2) plus jump intensity $\lambda \geq 0$, mean log-jump ν , and log-jump volatility $\delta > 0$. At $\lambda = 0$ the jump term vanishes and (5) collapses to the Heston level (1), so Heston is the $\lambda = 0$ face of the Bates parameter space. The model is selected at run time by a flag; the Heston fit is the default. The Feller quantity (4) is built from the diffusive parameters alone, so jumps do not enter it; it is reported under both models.

1.3. Pricing operators. Fix a valuation with maturity $\tau > 0$ (in years), strike K , and type $w \in \{+1, -1\}$ (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (8)$$

Black price [Black, 1976, Black and Scholes, 1973]. With total standard deviation $\Sigma = \sigma\sqrt{\tau}$,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (9)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where $\mathcal{N}$ is the standard normal CDF and $F = F(S, \tau)$.

Heston price. The Heston model price $C_H(\Phi; S, K, \tau, w)$ is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (10)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for $j \in \{1, 2\}$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (11)$$

and characteristic functions $f_j = \exp(C_j + D_j v_0 + iu \ln F)$ given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[(b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (12)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (13)$$

where $\mu_1 = \frac{1}{2}$, $\mu_2 = -\frac{1}{2}$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$. In code this integral is evaluated by the QuantLib [Ballabio and the QuantLib contributors] **AnalyticHestonEngine** (which uses a branch-stable formulation of (11)); equations (10)–(11) define the operator it realises.

Bates price. The Bates price $C_{\text{Bates}}(\Theta; S, K, \tau, w)$ has the same form as the Heston price (10)–(11), with each characteristic function f_j multiplied by the Merton lognormal jump factor of §1.2:

$$f_j^B(u) = f_j(u) e^{\tau \psi_j(u)}, \quad \psi_j(u) = \lambda \left(e^{iu_j \nu - \frac{1}{2} \delta^2 u_j^2} - 1 \right) - iu_j \lambda \xi, \quad (14)$$

with $u_2 = u$, $u_1 = u - i$ (the same measure shift that distinguishes f_1 from f_2), and $\xi = e^{\nu + \delta^2/2} - 1$ the compensator (6). The exponent ψ_j is the Merton log-jump cumulant, compensated by $-iu_j \lambda \xi$ so the drift correction of (5) carries through. At $\lambda = 0$ the factor is 1 and C_{Bates} reduces to C_H . In code the integral is evaluated by the QuantLib **BatesEngine**, the jump-augmented counterpart of the **AnalyticHestonEngine**.

Implied volatility. The Black-implied volatility of a price C is the inverse of (9) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (15)$$

computed numerically (QuantLib **blackFormulaImpliedStdDev** on the forward (8), returning Σ , then $\sigma = \Sigma/\sqrt{\tau}$). This operator defines the IV-space residuals of §5.

2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day d , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by i , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (16)$$

being respectively the quote timestamp, strike, type $w_i \in \{+1, -1\}$, traded size, trade price, trade-implied volatility, the underlying level *at trade time* t_i , and the days to maturity. Maturities enter the engine as τ_i in years under Actual/365, i.e. $T_i = \tau_i/365$.

The day is rate-stamped with $r = r(d)$ and $g = g(d)$, the most recent risk-free and dividend rates on or before d (a flat-forward curve for each; under flat curves the evaluation date is immaterial to T_i , so it is set to the QuantLib “today”).

Pre-filter. We keep only economically usable, Heston-appropriate quotes. Write the ratio moneyness $\varrho_i = e^{-|\ln(K_i/S_i)|} \in (0, 1]$ (it equals 1 at the money and falls as the strike moves out). The kept set is

$$\mathcal{D} = \{i : \sigma_i > 0, \tau_{\min} \leq \tau_i \leq \tau_{\max}, \underline{m} < \varrho_i < \bar{m}\}. \quad (17)$$

The maturity floor $\tau_{\min}$ (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive η, κ to Feller-violating extremes that pollute the pooled fit; the ceiling $\tau_{\max}$ drops thin, stale long tenors. The moneyness band keeps only out-of-the-money quotes: the cutoff $\bar{m}$ removes the near-ATM rows, and the floor $\underline{m}$ removes the deep-OTM lottery-ticket tail ($\varrho_i < \underline{m}$, roughly $|\ln(K_i/S_i)| > 0.51$), whose extreme prices peg the fit to its bounds. Flooring that tail recovers acceptance and tightens the fit while keeping the full tradeable wing.

3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot S , yet the trades in $\mathcal{D}$ occur across an intraday range of underlying levels $\{S_i\}$. We reconcile the two with one reference spot and a moneyness re-striking.

Reference spot. Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (18)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (19)$$

A flagged day ($R > R_{\max}$) is still calibrated and written, but the normalisation below is recorded as strained (its S_{ref} should be treated with suspicion).

Moneyness normalisation and grid snap. Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width Δ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (20)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in K/S rather than in the absolute strike. Re-striking

by (20) therefore re-centres the whole day onto S_{ref} so that trades executed at different S_i but the *same* moneyness share a common surface column. The grid snap collapses the continuum of K_i^* values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract's *original* (S_i, K_i) , since the calibrated parameters are spot-independent.

4. Surface construction

From the normalised trades we build a partial $K^* \times \tau$ implied-volatility surface, selecting the most liquid, nearest-the-money quotes.

Maturity selection. For each distinct maturity τ present in $\mathcal{D}$, let $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$ be its traded volume. Keep the N_t highest-volume maturities,

$$\mathcal{T} = \text{top-}N_t \text{ maturities by } V(\tau). \quad (21)$$

Strike selection (per maturity, per wing). Within each kept maturity $\tau \in \mathcal{T}$, separate calls ($w_i = +1$, OTM with $K_i^* \gtrsim S_{\text{ref}}$) from puts ($w_i = -1$, OTM with $K_i^* \lesssim S_{\text{ref}}$) and keep the n_k nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{n_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{n_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (22)$$

Pivot to a surface. For each retained cell (k, τ) with $k \in \mathcal{K}_\tau$, set the surface entry to the implied volatility of the *highest-volume* trade in that cell, the one with the largest traded size,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (23)$$

and leave it missing otherwise. When several trades tie at the maximum size the pick is arbitrary. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (24)$$

be the set of populated cells; these become the calibration targets (helpers).

Coverage requirement. To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (25)$$

A day failing (25) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

5. The optimization problem

The routine calibrates one parameter vector per day. Under the default Heston model this is Φ of (3) with $\dim \Phi = 5$; under Bates it is Θ of (7) with $\dim \Theta = 8$. The two are nested. They share their first $\dim \Phi$ coordinates, the diffusive parameters $(\theta, \kappa, \eta, \rho, v_0)$, and Θ appends the jump triple (λ, ν, δ) . Setting $\lambda = 0$ collapses the jump term (5) and recovers Φ , so Heston is the $\lambda = 0$ face of Bates. We write Ψ for the active vector, either Φ or Θ , and $C^{\text{mod}}(\Psi; \cdot)$ for the active model price: the Heston price C_H (10) when $\Psi = \Phi$, and the Bates price C_{Bates} (14) when $\Psi = \Theta$. Everything below is written in Ψ and holds for either model. The only place the model enters is the box dimension $\dim \Psi$ (26) and, through it, the bounds tables.

Let $\mathcal{N}$ index the targets with market data $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$, all priced at the single spot S_{ref} . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^{\dim \Psi} [\ell_j, h_j] \subset \mathbb{R}^{\dim \Psi}, \quad (26)$$

with (ℓ, h) in the `model.params()` order of Table 2: the five Heston coordinates, extended by the three jump bounds when $\Psi = \Theta$. The λ floor is exactly 0, so the Bates box contains the Heston model on its $\lambda = 0$ face.

5.1. Residuals. Two residual systems are defined: an IV-space system and a price-space system. The in-engine objective is configurable. It defaults to the IV-space system (30) below, and can instead be set to the price-space system (27). The choice changes only what each restart minimises. Selection (34) and the acceptance gate (36) always run on the IV-space error J_{iv} , so under the default objective the optimizer minimises the very error that then selects and gates.

Price-space (alternative in-engine objective). For each target the market value is the Black price at the quoted vol, $C_n^{\text{mkt}} = C_{\text{B}}(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$, and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Psi) = \frac{C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (27)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Psi) = \left(\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Psi)^2 \right)^{1/2}. \quad (28)$$

The per-target residual $\varepsilon_n^{\text{px}}$ is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt [Marquardt, 1963] is a sum-of-squares method, so it drives the squared residual norm $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Psi)^2$ to a local minimum (equivalently J_{px} , with which it shares a minimiser).

IV-space (default in-engine objective; selection and gate). Invert each model price back to a Black vol via (15) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Psi) &= \mathcal{I}(C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Psi) &= \sigma_n^{\text{mod}}(\Psi) - \hat{\sigma}_n, \end{aligned} \quad (29)$$

with objective

$$J_{\text{iv}}(\Psi) = \left(\frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Psi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Psi) \text{ inverts finitely}\}. \quad (30)$$

J_{iv} is measured in volatility points; it is the natural surface-fit metric and, unlike J_{px} , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

5.2. Multi-start local optimization. A single start into a non-convex problem is fragile [Mikhailov and Nögel, 2003], so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right), \quad (31)$$

and define the seed set $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$ of $(v_0, \kappa, \theta, \eta, \rho)$ tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (32)$$

Under Bates each seed is extended by one fixed jump triple $(\lambda, \nu, \delta) = (0.1, -0.1, 0.1)$, so every restart starts from almost no jumps and grows them only if they tighten the fit; the restart count is unchanged. For each seed $s \in \mathcal{S}$ we solve the box-constrained nonlinear least-squares problem

$$\hat{\Psi}_s = \arg \min_{\Psi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{obj}}(\Psi)^2 \quad (\text{LM, started at } \Psi^{(0)} = s), \quad (33)$$

using Levenberg–Marquardt under the box constraint $\mathcal{B}$ (`QuantLib NonhomogeneousBoundaryConstraint`). The per-target residual $\varepsilon_n^{\text{obj}}$ is the configured in-engine objective: by default the IV-space residual $\varepsilon_n^{\text{iv}}$ (29), optionally the price-space residual $\varepsilon_n^{\text{px}}$ (27). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Psi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Psi}_s). \quad (34)$$

Ranking by J_{iv} (not J_{px}) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

6. Acceptance gate

Ψ^* is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction δ , be

$$\text{PEGGED}(\Psi) = \bigvee_{j=1}^{\dim \Phi} \left[(\Psi_j - \ell_j) < \delta (h_j - \ell_j) \quad \vee \quad (h_j - \Psi_j) < \delta (h_j - \ell_j) \right]. \quad (35)$$

The disjunction runs over the $\dim \Phi$ diffusive coordinates for both models. Under Bates the three jump coordinates (ν, δ, λ) are exempt from the predicate: $\lambda \rightarrow 0$ is a legitimate collapse to Heston rather than the optimizer hitting a wall, and when $\lambda \approx 0$ the jump shape (ν, δ) is unidentified and may rest on a bound without meaning. The jump triple is still reported. The Feller quantity (4) likewise stays the diffusive quantity under Bates and never gates. The day is accepted iff

$$\boxed{J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Psi^*)}. \quad (36)$$

The first clause is the IV-space gate ($\varepsilon_{\text{iv}} \approx 2$ vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of J_{iv} . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

Why an IV-space gate. An earlier version gated on J_{px} directly. On a pooled day rich in deep-OTM wings, J_{px} is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with J_{iv} aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is ≈ 0.8 – 1.4 vol points, and the residual rejections are boundary-pegged κ or ρ (a separate identification lever), not fit-quality failures.

7. Output and diagnostics

An accepted day yields one parameter row keyed by date,

$$(d, S_{\text{ref}}, r, g, \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*, \mathcal{F}(\Psi^*), J_{\text{iv}}, J_{\text{px}}, \dots, R, \text{HIGH_MOVE}), \quad (37)$$

together with coverage counts ($|\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|$) and total volume. Under Bates the row appends the three jump parameters ($\lambda^*, \nu^*, \delta^*$) after v_0^* , and the repriced diagnostics carry a Bates price column in place of the Heston one; everything else is identical. Outputs are routed per model and objective, so the Heston and Bates rows never share a file. Because the calibration is once-daily, these rows accumulate into a single file, regenerated each run from the accepted days. For repricing diagnostics each retained contract is valued under Ψ^* at its *original* (S_i, K_i) , not (S_{ref}, K^*) , since the structural parameters are spot-independent; this gives an honest comparison against the actual trade price P_i .

8. Constants

Symbol	Code name	Value	Role
$\tau_{\min}$	MIN_DTM	14 days	minimum maturity
$\tau_{\max}$	MAX_DTM	730 days	maximum maturity
$\underline{m}$	OTM_MONEYNESSE_FLOOR	0.6	OTM ratio-moneyness band
$\bar{m}$	OTM_MONEYNESSE_CUTOFF	0.98	OTM ratio-moneyness band
$R_{\max}$	MAX_MOVE_PCT	0.03	intraday-range flag threshold
Δ	STRIKE_GRID	5.0	SPX strike lattice for K^* snap
N_t	MAX_NT	20	maturities kept (by volume)
n_k	MAX_NK	40	strikes kept per wing
M_t	MIN_MATS	3	minimum maturities in surface
M_k	MIN_STRIKES	5	minimum strikes in surface
M_{cell}	MIN_CELLS	12	minimum populated cells $ \mathcal{N} $
ε_{iv}	IV_RMSE_ACCEPT	0.02	IV-space acceptance gate (vol pts)
δ	BOUND_TOL	10^{-3}	boundary-pegging tolerance fraction

TABLE 1. Surface, selection, and gate constants.

	θ	κ	η	ρ	v_0		λ	ν	δ
lower ℓ	10^{-4}	10^{-2}	10^{-2}	-0.999	10^{-4}	lower ℓ	0.0	-0.5	10^{-3}
upper h	1.0	20.0	2.0	0.5	1.0	upper h	5.0	0.2	0.5

TABLE 2. Box bounds $\mathcal{B}$ (26), in the `model.params()` order. Left: the five Heston coordinates $(\theta, \kappa, \eta, \rho, v_0)$; the ρ upper bound is kept slightly positive (equity leverage implies $\rho < 0$) and η is capped at 2.0 (SPX vol-of-vol ~ 0.3 – 1.2). Right: the Bates jump bounds (λ, ν, δ) appended when $\Psi = \Theta$; the λ floor is exactly 0 (the fit may collapse to pure Heston) and ν allows more negative room since equity jumps skew down.

9. Empirical findings

All findings in this section are for the Heston model; a full-sample Bates run is left to later work (§9.2). We ran the routine over the full CBOE SPX trade history available to us, 3,217 trading days from 3 January 2012 to 15 October 2024. The figures below are for the default IV-space objective (30). The price-space objective (28) is the configurable alternative, discussed at the end of the section. Of these days, 1,631 (50.7%) passed the acceptance gate (36) and contribute a parameter row; the 1,586 rejected days are logged with their cause. The rejection split is dominated by boundary pegging: 1,467 days (92.5% of rejections) are rejected for a pegged parameter, 116 for an IV-space miss ($J_{\text{iv}} > \varepsilon_{\text{iv}}$), and 3 for having no usable trades. Pegging (almost always $\kappa \rightarrow 20$ or $\rho \rightarrow -0.999$ hitting the box wall) is therefore the binding constraint on the accept rate, not surface quality.

Fit quality. On the accepted set the surface fit is excellent. The IV-space error (30) has a median of 0.80 vol points (mean 0.84, 90th percentile 1.15, maximum 1.99), so J_{iv} sits inside the 2-vol-point gate on every accepted day and the gate never binds in practice. Acceptance is governed by the no-pegging clause alone. An independent repricing check (each contract valued under Φ^* at its *original* (S_i, K_i) and compared to the trade price) gives a per-day median relative price error of 11.4% with a 90th percentile of 17.4%. Those figures are inflated by the deep-OTM wings, whose near-zero prices make any absolute miss a large relative one. This is exactly the effect that motivated gating in IV space rather than price space (§6). A separate validator clears 1,623 of 1,631 accepted days (99.5%); the 8 hard failures are repricing-fit outliers, not parameter pathologies.

Parameter levels. The accepted parameters are economically sensible and stable across the 2012–2024 sample. The leverage correlation is consistently and strongly negative, median $\rho = -0.74$ (10th–90th percentile $[-0.81, -0.61]$), as equity-index smiles require. Long-run variance median $\theta = 0.067$ ($\approx 25.8\%$ annualised vol) and initial variance median $v_0 = 0.026$ ($\approx 16.2\%$) bracket realistic SPX levels, with v_0 the lower of the two on the typical day. Mean reversion median $\kappa = 2.37$ sits in the usual SPX range. The vol-of-vol is high, median $\eta = 1.34$ (10th–90th percentile $[0.79, 1.84]$). Because the gate rejects any boundary-pegged fit, the accepted set is pegging-free by construction: no accepted day carries ρ below -0.99 or reaches the κ ceiling. A clean interior set is what survives the gate, not evidence that pegging is solved on the underlying population.

Feller condition is the standout caveat. The gate does not test Feller, and these fits violate it without exception: $\mathcal{F}(\Phi^*) = 2\kappa\theta - \eta^2 < 0$ on all 1,631 accepted days, with median $\mathcal{F} = -1.44$. The driver of the violation is the large vol-of-vol relative to $\kappa\theta$. η exceeds 1.5 on 596 days (36.5%) and approaches its 2.0 cap (max 1.997). A pinned v_0 or a degenerate κ are rare, so among the validator’s soft flags Feller violation and an elevated η are by far the most common. Together they mark these fits as ones whose variance path can reach zero, even where the surface fit itself is tight. The wide surface coverage is what lifts η here: putting the full OTM wings into the fit demands more vol-of-vol than a near-money fit, so both η and the Feller violation run higher than under the narrower near-money calibration of earlier drafts.

Comparison with the earlier price-space baseline. An earlier draft reported the price-space objective on a narrower surface (near-money strikes only, a higher maturity floor) as the default. That run accepted 1,713 of 3,215 days (53.3%) with a lower vol-of-vol (median $\eta \approx 0.89$, $\eta > 1.5$ on 8.5% of days) and Feller violated on 99.3%. Two things changed the picture here. The default objective is now the IV-space residual, and the surface coverage was widened to carry the full tradeable wing and longer maturities, with a deep-OTM floor that drops the lottery-ticket tail. The widening is the larger effect. It removes what looked like a systematic wing underfit: with the wings in the fit, the per- $|\ln(K/S)|$ residual is small and mixed in sign (overall RMSE 1.1 vol points), so the near-money-only calibration was extrapolating, not capturing an in-sample bias. The same widening raises η and pushes Feller violation to every accepted day. The accept rate is little changed (50.7% against 53.3%), and pegging stays the dominant rejection cause under either setup.

Price-space as the alternative objective. Switching the in-engine objective back to price-space (28) changes only what each restart minimises, not how a day is selected or gated (§5). The IV-space default weights the deep-OTM wings evenly in vol points, whereas the price objective discounts them by their small prices. One consequence is specific to the IV-space default: it no longer controls relative price, so the returned price RMSE is large on the cheap wing cells (maximum $J_{\text{px}} = 120.8$, with 656 days above 0.2). Those days are tight in vol points yet far off in relative price on a near-worthless wing cell, which is why the gate runs in IV space.

9.1. Partial conclusions. The objective is not the binding choice. Pegging is a structural degeneracy in the (κ, ρ, η) skew subspace, and it survives a change of residual. The pooled SPX surface demands more skew than single-factor Heston can supply without driving one of κ or ρ to a corner, and the optimiser picks whichever corner. Acceptance therefore turns on the no-pegging clause, not on fit quality or on which residual is minimised. The levers that would move it are the identification fixes: anchoring κ , a soft Feller penalty, and liquidity weighting.

9.2. Avenues for improvement. Acceptance is near one half of attempted days, and Feller violation is now universal on the accepted set. The maturity floor has already been raised to $\tau_{\min} = 14$ days and an objective wing-weighting has been wired but tested null; neither moved acceptance. The remaining levers attack the (κ, ρ, η) degeneracy at its source. Fixing or anchoring κ from a cross-day-robust estimate would break the $\kappa \leftrightarrow \rho$ trade-off so ρ stops pegging. A soft Feller penalty in the restart-selection criterion, with a revisit of the η cap, would curb the now-universal Feller violation. Warm-starting each day from the previous day's accepted parameters would stabilise the search. The Levenberg–Marquardt end criteria have not been tuned and are a further candidate.

A full-sample Bates calibration is the other open item. The engine, bounds, and routing are in place (§1.2, Table 2), and the jumps give the model a direct way to carry the tail the single-factor vol-of-vol now overfits. Whether that relieves the (κ, ρ, η) degeneracy and the η -driven Feller violation across the full 2012–2024 sample is the question a full Bates run would settle.

Rough volatility is the natural next paradigm, but a separate one. Empirical realised variance is well described by a fractional process with Hurst exponent

$H \approx 0.1$ [Gatheral et al., 2018], far from the $H = \frac{1}{2}$ that the diffusive Heston and Bates dynamics assume. A rough model fits the steep short-maturity skew that single-factor Heston cannot reach without driving η to its cap, so it speaks directly to the degeneracy and Feller findings above. It does not, however, fit the routine of this paper. The dynamics are non-Markovian, so the finite-dimensional box, the seed grid, and the Feller quantity have no direct analogue. There is no QuantLib engine for it. The fastest tractable variant, rough Heston, prices through a fractional Riccati equation [El Euch and Rosenbaum, 2019] that is far heavier per evaluation than the closed-form Heston integral, and rough Bergomi [Bayer et al., 2016] prices by Monte Carlo. Multi-start calibration over thousands of days is then out of reach without a surrogate pricer. There is also a tension with the present surface: the rough skew is sharpest at the shortest maturities, exactly the tenors the pre-filter drops at $\tau_{\min} = 14$ days (17). A fair test would re-admit the short end, build a rough-Heston pricer or a learned surrogate, and run on a pilot window rather than the full sample. We leave it to separate work.

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Appendix A. Example option prices

Heston model prices produced by the market¹ calibrated pricing operator:

$$C_H(\Phi^*; S, K, \tau, w) \quad (10) \quad (36).$$

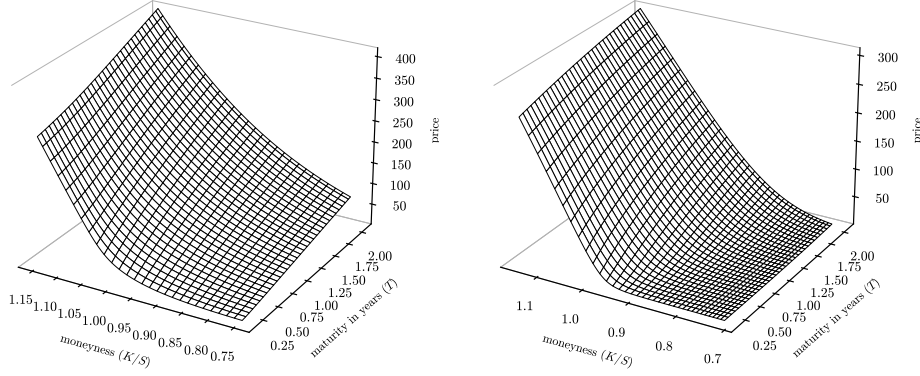


FIGURE 1. Heston option prices for S_{ref} 1957.9267 on October 24, 2014 with $\Phi^* = (0.0538, 3.4097, 1.2288, -0.7475, 0.024)$: puts wing (left) and calls wing (right).

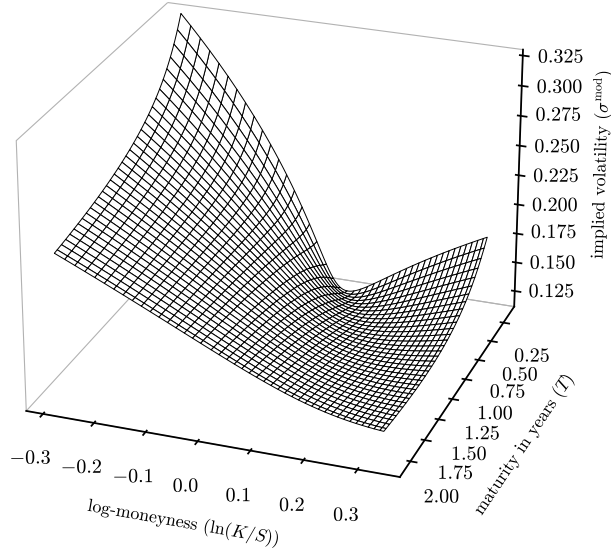


FIGURE 2. *Out of the money* implied volatilities from Figure 1

¹The fit uses 656 calibration cells across 15 maturities and 140 strikes, backed by 240,081 contracts of traded volume. The reference spot was $S_{\text{ref}} = 1957.9267$, priced under a risk-free rate of 0.09% and a dividend rate of 1.97%. The intraday spot range was 0.97%. Fit quality is 0.54 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.1371. The Feller condition violates at this calibration, with $2\kappa\theta - \eta^2 = -1.1428$.

Bates model prices produced by the market² calibrated pricing operator:
 $C_{\text{Bates}}(\Theta^*; S, K, \tau, w)$ (14) (36).

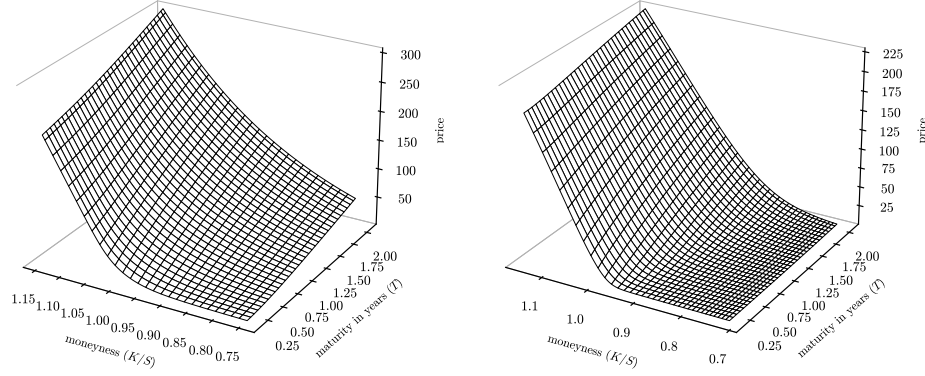


FIGURE 3. Bates option prices for S_{ref} 1500.3753 on January 25, 2013 with $\Theta^* = (0.0615, 0.5521, 0.3767, -0.7631, 0.0103, 0.0723, -0.2699, 0.3295)$: puts wing (left) and calls wing (right).

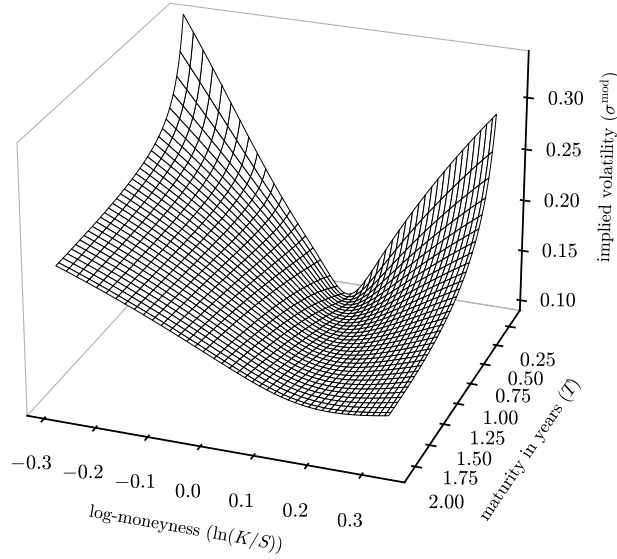


FIGURE 4. *Out of the money* implied volatilities from Figure 3

²The fit uses 418 calibration cells across 13 maturities and 123 strikes, backed by 280,247 contracts of traded volume. The reference spot was $S_{\text{ref}} = 1500.3753$, priced under a risk-free rate of 0.14% and a dividend rate of 2.13%. The intraday spot range was 0.56%. Fit quality is 0.36 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.1217. The Feller condition violates at this calibration, with $2\kappa\theta - \eta^2 = -0.074$.

Appendix B. Example fits for the Heston model

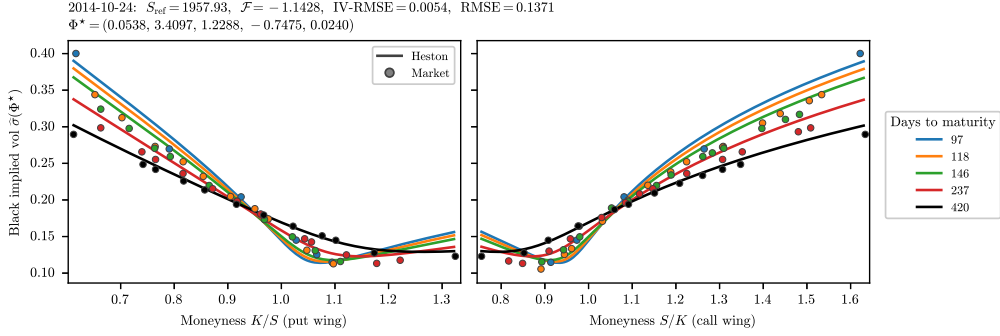


FIGURE 5. Heston implied-volatility smiles for October 24, 2014: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

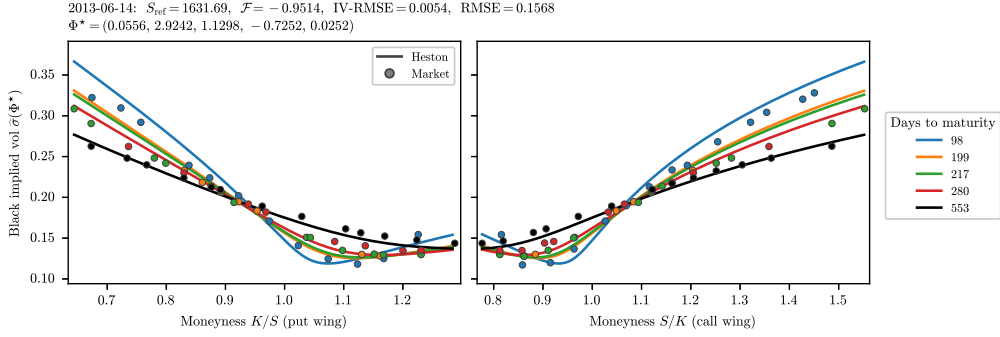


FIGURE 6. Heston implied-volatility smiles for June 14, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

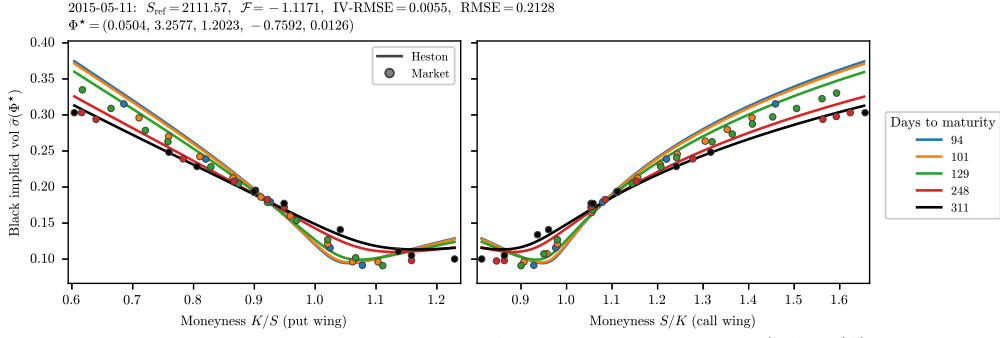


FIGURE 7. Heston implied-volatility smiles for May 11, 2015: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

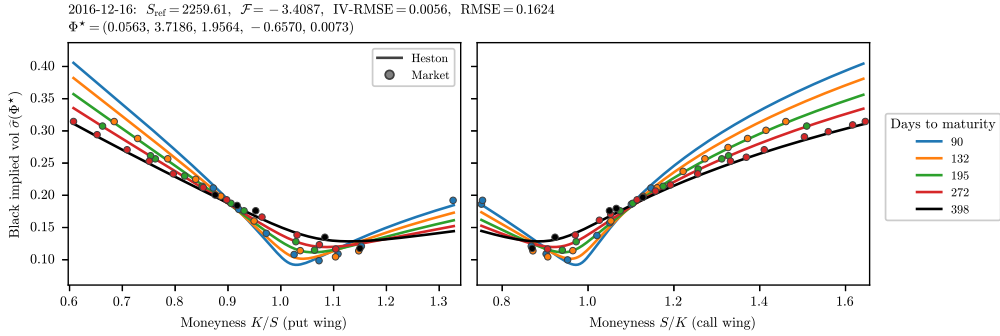


FIGURE 8. Heston implied-volatility smiles for December 16, 2016: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2016-08-23: $S_{ref} = 2190.37$, $\mathcal{F} = -2.9476$, IV-RMSE = 0.0056, RMSE = 0.1897
 $\Phi^* = (0.0571, 3.5052, 1.8298, -0.7352, 0.0065)$

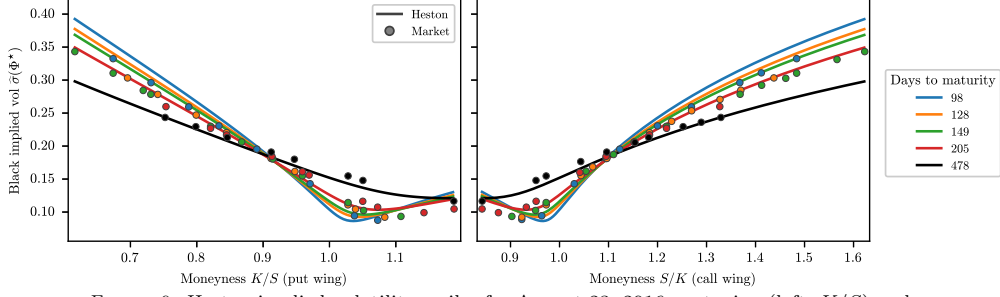


FIGURE 9. Heston implied-volatility smiles for August 23, 2016: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2017-01-27: $S_{ref} = 2293.84$, $\mathcal{F} = -3.2560$, IV-RMSE = 0.0057, RMSE = 0.1973
 $\Phi^* = (0.0543, 2.9068, 1.8899, -0.6046, 0.0048)$

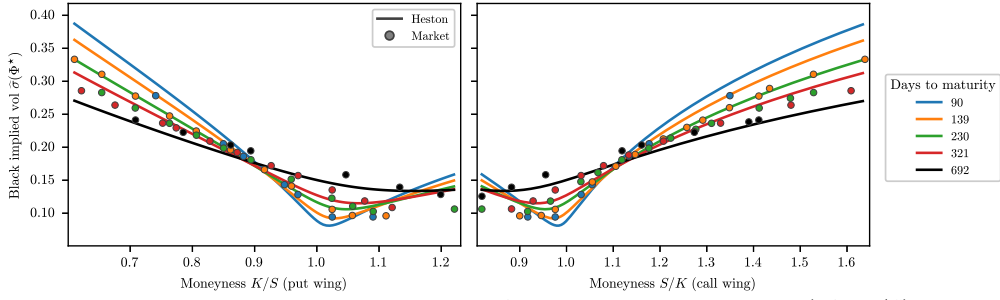


FIGURE 10. Heston implied-volatility smiles for January 27, 2017: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2015-05-22: $S_{ref} = 2129.02$, $\mathcal{F} = -1.4704$, IV-RMSE = 0.0057, RMSE = 0.2020
 $\Phi^* = (0.0454, 3.6931, 1.3437, -0.7294, 0.0081)$

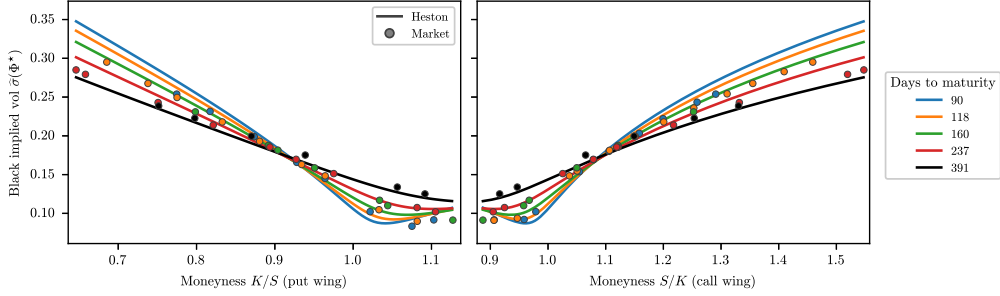


FIGURE 11. Heston implied-volatility smiles for May 22, 2015: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2015-05-28: $S_{ref} = 2117.65$, $\mathcal{F} = -1.7266$, IV-RMSE = 0.0058, RMSE = 0.2016
 $\Phi^* = (0.0434, 4.1796, 1.4455, -0.7511, 0.0134)$

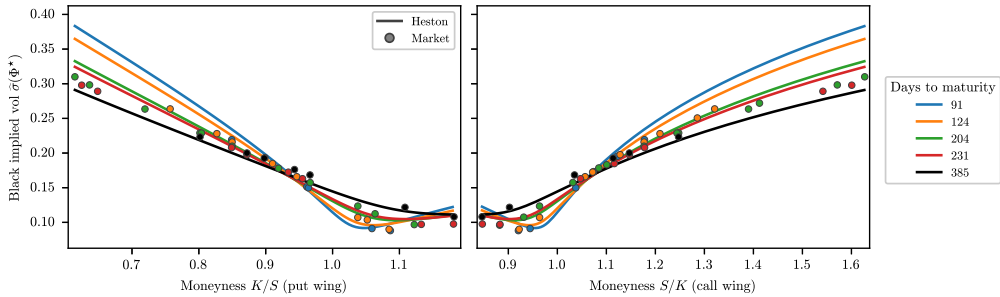
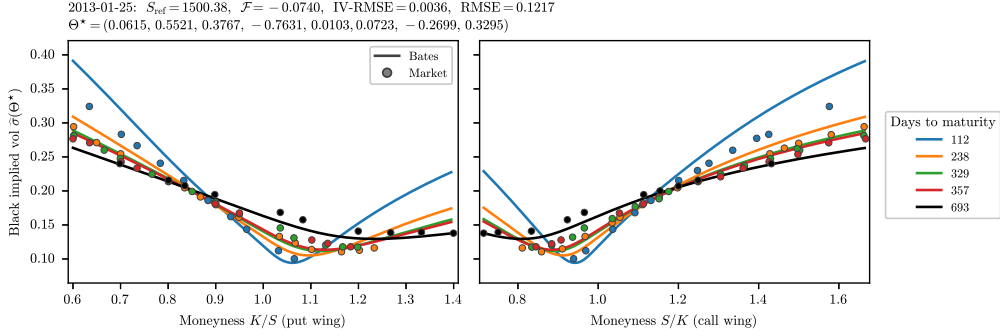
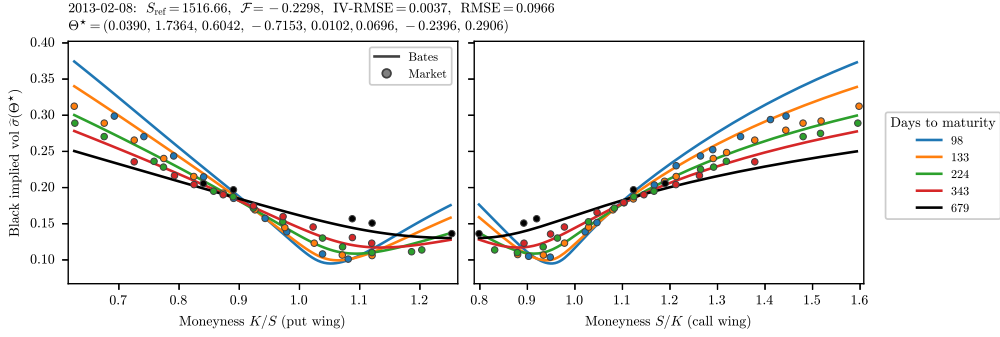
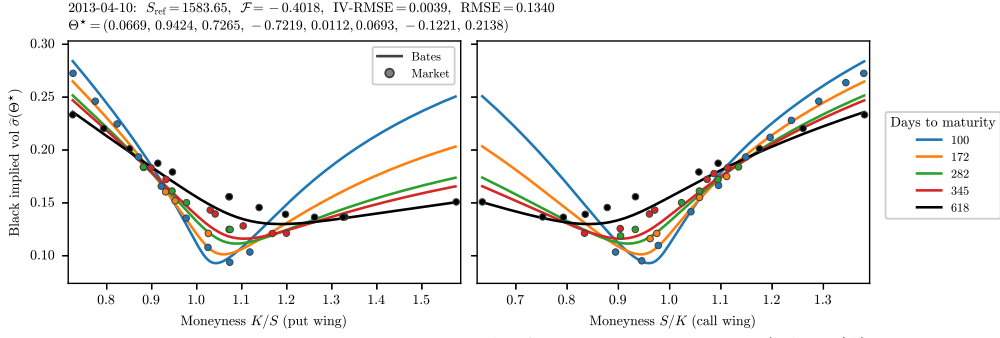
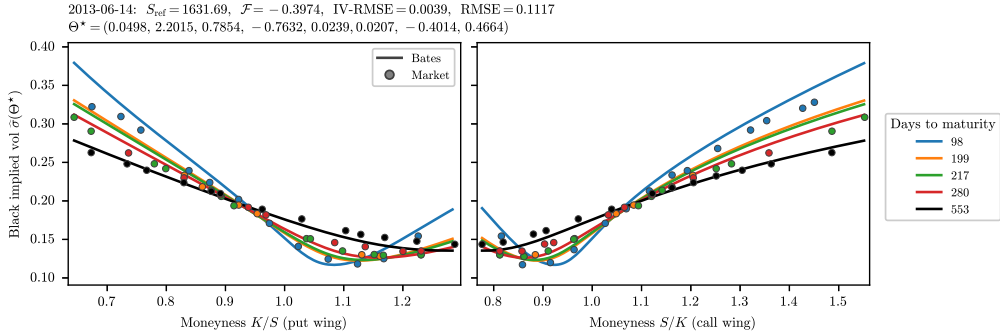


FIGURE 12. Heston implied-volatility smiles for May 28, 2015: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

Appendix C. Example fits for the Bates model

FIGURE 13. Bates implied-volatility smiles for January 25, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.FIGURE 14. Bates implied-volatility smiles for February 08, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.FIGURE 15. Bates implied-volatility smiles for April 10, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.FIGURE 16. Bates implied-volatility smiles for June 14, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-02-11: $S_{ref} = 1516.36$, $\mathcal{F} = -0.5837$, IV-RMSE = 0.0039, RMSE = 0.1062
 $\Theta^* = (0.0424, 2.0805, 0.8718, -0.6801, 0.0104, 0.0927, -0.1469, 0.2154)$

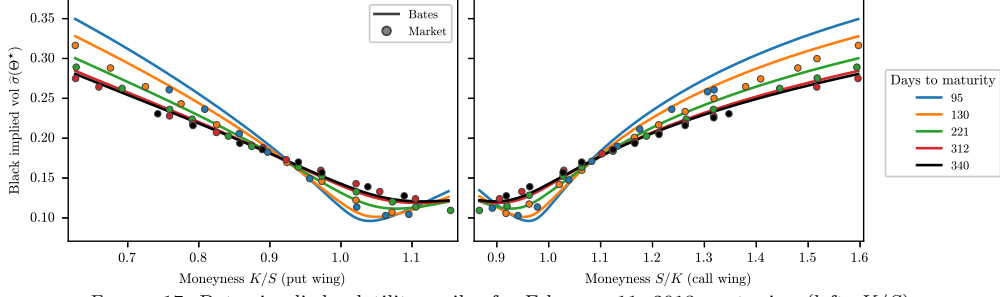


FIGURE 17. Bates implied-volatility smiles for February 11, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-05-06: $S_{ref} = 1617.28$, $\mathcal{F} = -0.2013$, IV-RMSE = 0.0040, RMSE = 0.1424
 $\Theta^* = (0.0505, 1.0074, 0.5505, -0.6750, 0.0125, 0.0244, -0.3850, 0.4548)$

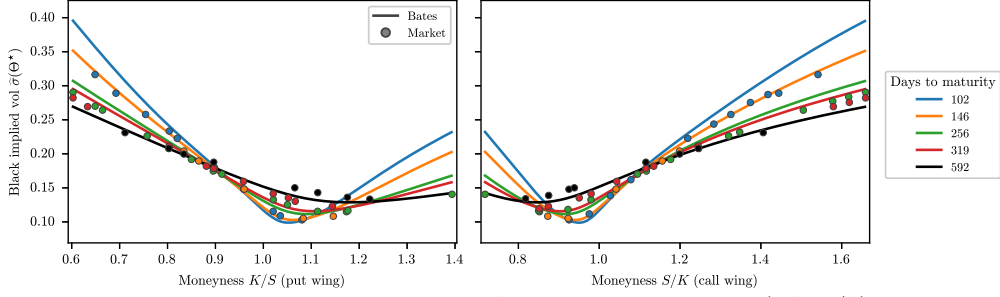


FIGURE 18. Bates implied-volatility smiles for May 06, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-07-09: $S_{ref} = 1649.59$, $\mathcal{F} = -0.1190$, IV-RMSE = 0.0040, RMSE = 0.1250
 $\Theta^* = (0.0603, 0.7789, 0.4614, -0.7368, 0.0169, 0.0285, -0.3997, 0.5000)$

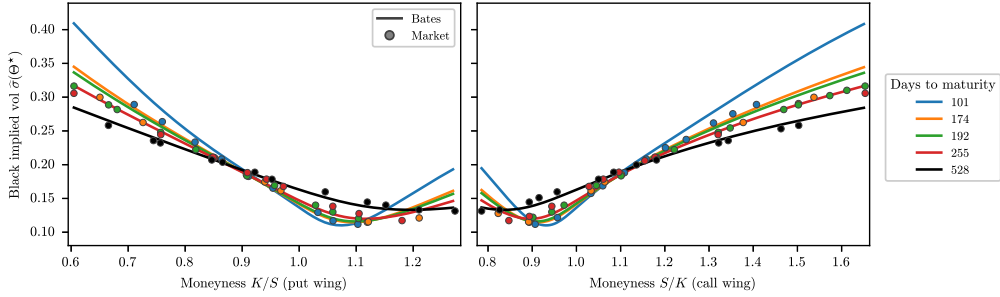


FIGURE 19. Bates implied-volatility smiles for July 09, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

2013-06-18: $S_{ref} = 1648.36$, $\mathcal{F} = -0.5224$, IV-RMSE = 0.0040, RMSE = 0.1202
 $\Theta^* = (0.0513, 1.9739, 0.8513, -0.7539, 0.0240, 0.0393, -0.2191, 0.3321)$

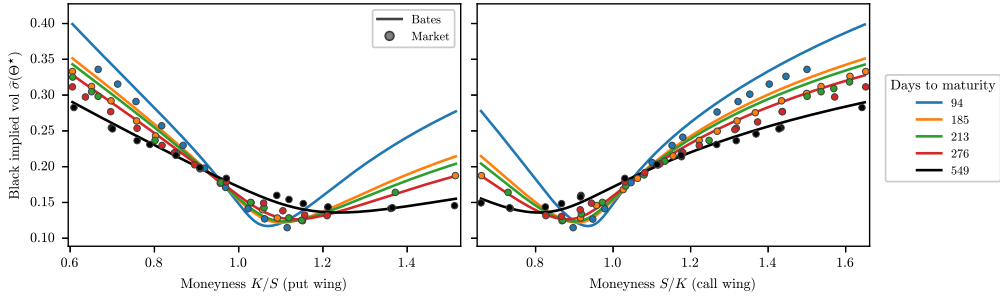


FIGURE 20. Bates implied-volatility smiles for June 18, 2013: put wing (left, K/S) and call wing (right, S/K), with market trades scattered.

Appendix D. Algorithm

Algorithm 1 states the per-day routine in the active vector Ψ ; the driver applies it to every trading day and accumulates accepted rows into a single file (§7). Heston is the default ($\Psi = \Phi$). The Bates fit is the same algorithm with $\Psi = \Theta$: the engine is the **BatesEngine** and the box $\mathcal{B}$ is the dim Θ -dimensional one, while the pegging test (35) still ranges over the dim Φ diffusive coordinates only.

Algorithm 1 Per-day calibration

Require: day d of OTM trades; rates $r(d), g(d)$; constants of Table 1

- 1: $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$ (pre-filter (17))
- 2: **if** $r(d)$ or $g(d)$ undefined on/before d **then**
- 3: **return** $\emptyset$ (no rate)
- 4: **end if**
- 5: $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$;
 $R \leftarrow \max_i S_i / \min_i S_i - 1$;
 $\text{HIGH_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6: $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$ with $m_i = K_i / S_i$ (re-strike (20))
- 7: $\mathcal{T} \leftarrow$ top- N_t maturities by volume; build $\mathcal{K}_\tau$ (n_k per wing);
 pivot to $\{\hat{\sigma}_{k,\tau}\}$, cells $\mathcal{N}$
- 8: **if** coverage (25) fails **then**
- 9: clear stale diagnostics; **return** $\emptyset$
- 10: **end if**
- 11: $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$; build seeds $\mathcal{S}$ (32)
- 12: **for** $s \in \mathcal{S}$ **do**
- 13: $\hat{\Psi}_s \leftarrow$ LM minimiser of $\sum_n (\varepsilon_n^{\text{obj}})^2$ on $\mathcal{B}$ from s (33)
- 14: compute $J_{\text{iv}}(\hat{\Psi}_s)$ (30)
- 15: **end for**
- 16: $\Psi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Psi}_s)$ (select (34))
- 17: **if** $J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}}$ **and** $\neg \text{PEGGED}(\Psi^*)$ **then**
- 18: write repriced diagnostics; **return** $\text{row}(d, S_{\text{ref}}, r, g, \Psi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20: clear stale diagnostics; **return** $\emptyset$ (rejected)
- 21: **end if**
